Test Series Code

12

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 12/04/2026

ANSWER KEY

1.	(4)	46.	(1)	91.	(2)	136.	(3)
2.	(3)	47.	(4)	92.	(4)	137.	(3)
3.	(2)	48.	(3)	93.	(1)	138.	(2)
4.	(2)	49.	(1)	94.	(1)	139.	(3)
5.	(2)	50.	(3)	95.	(1)	140.	(1)
6.	(3)	51.	(2)	96.	(2)	141.	(3)
7.	(2)	52.	(4)	97.	(2)	142.	(1)
8.	(2)	53.	(4)	98.	(3)	143.	(2)
9.	(4)	54.	(1)	99.	(4)	144.	(3)
10.	(3)	55.	(1)	100.	(4)	145.	(3)
11.	(3)	56.	(4)	101.	(3)	146.	(3)
12.	(2)	57.	(3)	102.	(2)	147.	(2)
13.	(1)	58.	(2)	103.	(4)	148.	(3)
14.	(2)	59.	(4)	104.	(3)	149.	(1)
15.	(3)	60.	(2)	105.	(1)	150.	(3)
16.	(3)	61.	(3)	106.	(2)	151.	(3)
17.	(1)	62.	(1)	107.	(1)	152.	(2)
18.	(2)	63.	(2)	108.	(3)	153.	(3)
19.	(4)	64.	(1)	109.	(1)	154.	(4)
20.	(3)	65.	(4)	110.	(2)	155.	(2)
21.	(1)	66.	(1)	111.	(1)	156.	(4)
22.	(3)	67.	(2)	112.	(3)	157.	(2)
23.	(4)	68.	(1)	113.	(4)	158.	(3)
24.	(1)	69.	(1)	114.	(2)	159.	(3)
25.	(4)	70.	(3)	115.	(1)	160.	(3)
26.	(2)	71.	(4)	116.	(4)	161.	(3)
27.	(3)	72.	(1)	117.	(1)	162.	(3)
28.	(2)	73.	(4)	118.	(4)	163.	(3)
29.	(1)	74.	(1)	119.	(2)	164.	(4)
30.	(1)	75.	(3)	120.	(2)	165.	(3)
31.	(1)	76.	(2)	121.	(3)	166.	(4)
32.	(3)	77.	(3)	122.	(4)	167.	(2)
33.	(3)	78.	(1)	123.	(3)	168.	(3)
34.	(4)	79.	(4)	124.	(2)	169.	(2)
35.	(4)	80.	(3)	125.	(1)	170.	(3)
36.	(3)	81.	(4)	126.	(3)	171.	(3)
37.	(1)	82.	(3)	127.	(3)	172.	(2)
38.	(2)	83.	(1)	128.	(2)	173.	(1)
39.	(3)	84.	(3)	129.	(4)	174.	(4)
40.	(3)	85.	(2)	130.	(1)	175.	(4)
41.	(3)	86.	(2)	131.	(2)	176.	(1)
42.	(4)	87.	(4)	132.	(2)	177.	(1)
43.	(3)	88.	(2)	133.	(4)	178.	(3)
44.	(1)	89.	(2)	134.	(1)	179.	(2)
45.	(2)	90.	(2)	135.	(2)	180.	(2)

Hints & Solutions

1. (4)

AND: Output is 1 only when both A and B are 1 (Truth table: $0,0,0,1$). This is incorrect.

NAND: Output is 0 only when both A and B are 1 (Truth table: $1,1,1,0$). This is incorrect.

OR: Output is 1 if at least one input is 1 (Truth table: $0,1,1,1$). This is incorrect.

(NCERT Ed. 2025-26, 12th, Page No. 341)

2. (3)

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

$$\frac{1.5}{v}-\frac{1}{-3R}=\frac{1.5-1}{R}$$

$$\frac{1.5}{v}+\frac{1}{3R}=\frac{0.5}{R}$$

$$v=1.5\times 6R$$

$$v=9R$$

(NCERT Ed. 2025-26, 12th, Page No. 233)

3. (2)

$$\mu_r=\frac{B}{\mu_0 H}=\frac{\Phi}{A\mu_0 H}$$

$$\mu_r=\frac{10^{-5}}{\left(2\times10^{-4}\right)\times\left(4\pi\times10^{-7}\right)\times\left(\frac{10^5}{4\pi}\right)}$$

$$\mu_r=\frac{10^{-5}}{2\times10^{-4}\times10^{-2}}=\frac{10^{-5}}{2\times10^{-6}}=5$$

$$\chi=\mu_r-1=5-1=4$$

(NCERT Ed. 2025-26, 12th, Page No. 146)

4. (2)

$$v_e=\sqrt{\frac{2GM}{R}}.$$

For planet: $M'=2M$, $R'=R/2$.

$$v_p=\sqrt{\frac{2G(2M)}{R/2}}=\sqrt{4\frac{2GM}{R}}=2\sqrt{\frac{2GM}{R}}=2v_e.$$

(NCERT Ed. 2025-26, 11th, Page No. 136)

5. (2)

Cutting the magnet perpendicular to its length into two equal parts results in each piece having a magnetic moment $M'=\frac{M}{2}$.

When the two pieces are stacked with "like poles together," the total magnetic moment of the system becomes $M_{total}=M'+M'=\frac{M}{2}+\frac{M}{2}=M$.

Since the total magnetic moment M remains the same as the original magnet and the magnetic field B is constant, the potential energy $U=-MB\cos\theta$ remains U.

(NCERT Ed. 2025-26, 12th, Page No. 139)

6. (3)

1. Object X is Positive $(+)$: A glass rod rubbed with silk becomes positive. Since X repels it, X must also be positive.
2. Object Z is Negative $(-)$: A plastic ruler rubbed with hair becomes negative. Since Z repels it, Z must also be negative.
3. Object Y is Neutral: A charged object would repel one of the others. Only an electrically neutral object is attracted to both positive and negative charges (due to induction /polarization).

(NCERT Ed. 2025-26, 12th, Page No. 2)

7. (2)

$$q_{\text{inner}}=\frac{Q_1-Q_2}{2}$$

$$q_{\text{inner (Plate I)}}=\frac{Q-2Q}{2}=-\frac{Q}{2}$$

(NCERT Ed. 2025-26, 12th, Page No. 69)

8. (2)

Thermal conductivity depends on the nature of material of the wall.

(NCERT Ed. 2025-26, 11th, Page No. 215)

9. (4)

From the derivation of Kepler's Third Law (equating centripetal force to gravitational force), we get:

$$T^2=\left(\frac{4\pi^2}{GM}\right)R^3$$

- Here, M is the mass of the central body (the source of gravity).
- Statement 2 is True: The constant $K=\frac{4\pi^2}{GM}$ depends on the central mass.
- Statement 1 is False: For the Moon, the central body is Earth; for the Earth, the central body is the Sun. Since $M_{Earth}\neq M_{Sun}$, the constant K is different for the two systems.

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10. (3)

Formula: $K = \frac{1}{2}mv^2$

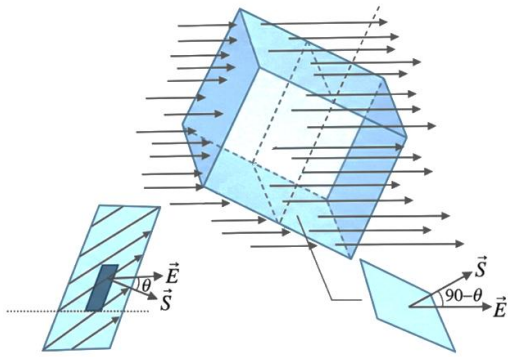
Percentage error formula:

$$\frac{\Delta K}{K} \times 100 = \frac{\Delta m}{m} \times 100 + 2\left(\frac{\Delta v}{v} \times 100\right)$$

Calculation: $2\% + 2(3\%) = 2 + 6 = 8\%$.

(NCERT Ed. 2025-26, 11th, Page No. 6)

11. (3)



total flux of electric field entering the cube in this position = flux through front tilted face + flux through bottom tilted face

$\phi_{total} = \phi_1 + \phi_2$

$= EL^2 (\cos\theta + \sin\theta)$

Express $(\cos\theta + \sin\theta)$ as a square root to match the choices:

$$\cos\theta + \sin\theta = \sqrt{(\cos\theta + \sin\theta)^2}$$

$$= \sqrt{\cos^2\theta + \sin^2\theta + 2\sin\theta\cos\theta}$$

$$= \sqrt{1 + \sin(2\theta)}$$

Therefore, the flux is $E \cdot l^2 \sqrt{1 + \sin(2\theta)}$

(NCERT Ed. 2025-26, 12th, Page No. 16)

12. (2)

For two projectiles launched simultaneously from the same ground level to collide in mid-air, their vertical heights must match at every instant.

Since gravity acts equally on both, their initial vertical velocities must be equal:

$u_{y1} = u_{y2}$

$u_1 \sin\theta_1 = u_2 \sin\theta_2$

(NCERT Ed. 2025-26, 11th, Page No. 38)

13. (1)

Energy of photon

$$E = \frac{12400}{\lambda(\text{in Å})} = \frac{12400}{2000} = 6.2 \text{ eV}.$$

Using Einstein's Photoelectric equation:

$K_{max} = E - \phi$

$K_{max} = 6.2 \text{ eV} - 4.2 \text{ eV} = 2.0 \text{ eV}.$

(NCERT Ed. 2025-26, 12th, Page No. 281)

14. (2)

For capacitors in series:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\frac{1}{C_{eq}} = \frac{1}{C} + \frac{1}{2C} + \frac{1}{3C}$$

$$\frac{1}{C_{eq}} = \frac{6+3+2}{6C} = \frac{11}{6C}$$

$$C_{eq} = \frac{6C}{11}.$$

(NCERT Ed. 2025-26, 12th, Page No. 72)

15. (3)

Work $W = \int \vec{F} \cdot d\vec{r} = \int (F_x dx + F_y dy)$

$$W = \int_0^2 3\,dx + \int_0^2 4\,dy = [3x]_0^2 + [4y]_0^2$$

$W = 3(2-0) + 4(2-0) = 6 + 8 = 14$ J.

(NCERT Ed. 2025-26, 11th, Page No. 74)

16. (3)

First Step (Stretching):

When a wire is stretched to n times its length, resistance becomes n^2R.

Here $n = 2$, so new resistance $R' = (2)^2 R = 4R$.

Second Step (Compression):

Now, the wire (with resistance $4R$) is compressed to $\frac{1}{4}$ of its length ($n = \frac{1}{4}$).

Final resistance $R_{final} = \left(\frac{1}{4}\right)^2 \times (4R)$.

$$R_{final} = \frac{1}{16} \times 4R = \frac{R}{4}$$

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17. (1)

Resonant angular frequency $\omega = \frac{1}{\sqrt{LC}}$

$$\omega = \frac{1}{\sqrt{2 \times 32 \times 10^{-6}}} = \frac{1}{\sqrt{64 \times 10^{-6}}}$$

$$= \frac{1}{8 \times 10^{-3}} = \frac{1000}{8} = 125 \text{ rad/s}.$$

Frequency $f = \frac{\omega}{2\pi} = \frac{125}{2 \times 3.14} \approx 20$ Hz.

(NCERT Ed. 2025-26, 12th, Page No. 189)

18. (2)

- Top (Same Size): A Plane mirror forms an image of the same size as the object.
- Middle (Smaller): A Convex mirror forms a diminished (smaller) image.
- Bottom (Same Size): A Plane mirror forms an image of the same size as the object.

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19. (4)

When a charged particle enters a magnetic field at an angle θ (where $0^\circ < \theta < 90^\circ$), the velocity component perpendicular to the field ($v\sin\theta$) provides circular motion, and the component parallel to the field ($v\cos\theta$) provides linear motion. The combined path is helical.

(NCERT Ed. 2025-26, 12th, Page No. 112)

20. (3)

Young's Modulus $Y = \frac{FL}{A\Delta L}$. Since F, L, A are same, $Y \propto \frac{1}{\Delta L}$.

Ratio $Y_1 : Y_2 : Y_3 = \frac{1}{x} : \frac{1}{2x} : \frac{1}{3x} = 1 : \frac{1}{2} : \frac{1}{3}$.

Multiplying by 6 (LCM), we get $6:3:2$.

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21. (1)

Using Bernoulli's equation for a horizontal pipe:

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

$$P + \frac{1}{2}\rho v^2 = P_2 + \frac{1}{2}\rho (2v)^2$$

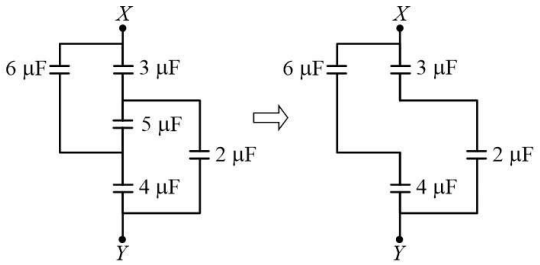
$$P + \frac{1}{2}\rho v^2 = P_2 + 2\rho v^2$$

$$P_2 = P + \frac{1}{2}\rho v^2 - 2\rho v^2 = P - \frac{3}{2}\rho v^2.$$

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22. (3)

Equivalent circuit may be drawn as follows:



As the structure is balanced Wheatstone bridge, so 5μF capacitor will draw no charge from battery and may be removed. Now the circuit is a simple circuit.

$$C_{eq} = \frac{4\times 6}{(6+4)} + \frac{3\times 2}{(3+2)} = \frac{18}{5}\mu F$$

(NCERT Ed. 2025-26, 12th, Page No. 73)

23. (4)

According to Stefan's Law, $E \propto T^4$.

New Temperature $T' = T + 0.5T = 1.5T = \frac{3}{2}T$.

New Energy $E' \propto (1.5T)^4 = 5.0625T^4$.

Percentage increase

$$= \frac{E' - E}{E} \times 100 = (5.0625 - 1) \times 100 = 406.25\%.$$

(NCERT Ed. 2025-26, 11th, Page No. 219)

24. (1)

Work $W = \int PdV = \int_V^{2V} (\alpha V) dV$.

$$W = \alpha[\frac{V^2}{2}]_V^{2V} = \frac{\alpha}{2}\left[(2V)^2 - V^2\right].$$

$$W = \frac{\alpha}{2}\left[4V^2 - V^2\right] = \frac{3}{2}\alpha V^2.$$

(NCERT Ed. 2025-26, 11th, Page No. 234)

25. (4)

$$\lambda = \frac{h}{\sqrt{2mqV}}.$$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}}.$$

For alpha particle: $m_\alpha = 4m_p$ and $q_\alpha = 2q_p$.

Ratio $= \sqrt{\frac{4m_p \cdot 2q_p}{m_p \cdot q_p}} = \sqrt{8} = 2\sqrt{2} : 1.$

(NCERT Ed. 2025-26, 12th, Page No. 285)

26. (2)

Efficiency $\eta = 1 - \frac{T_{sink}}{T_{source}}$.

$$0.5 = 1 - \frac{300}{T_{source}}.$$

$$\frac{300}{T_{source}} = 1 - 0.5 = 0.5$$

$$T_{source} = \frac{300}{0.5} = 600 \text{ K}$$

(NCERT Ed. 2025-26, 11th, Page No. 238)

27. (3)

$$v_{rms} \propto \sqrt{\frac{T}{M}}$$

Initially: $v \propto \sqrt{\frac{T}{32}}$ (Molecular mass of O_2 is 32).

Finally: $T' = 2T$ and gas dissociates to atomic oxygen (O),

so $M' = 16$.

$$v'_{rms} \propto \sqrt{\frac{2T}{16}}.$$

Ratio $\frac{v'}{v} = \sqrt{\frac{2T/16}{T/32}} = \sqrt{\frac{2\times 32}{16}} = \sqrt{4} = 2.$

$v' = 2v.$

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28. (2)

Total Energy $(E) = U + K = 8K + K = 9K.$

Therefore, $U = \frac{8}{9}E.$

Since $U \propto x^2$ and $E \propto A^2$:

$x^2 = \frac{8}{9}A^2$

$x = \sqrt{\frac{8}{9}}A = \frac{2\sqrt{2}}{3}A$

$x = \frac{2\sqrt{2}}{3} \times 30$

$x = 2\sqrt{2} \times 10$

$x = 20\sqrt{2}$ cm

(NCERT Ed. 2025-26, 11th, Page No. 269)

29. (1)

$B = \frac{\mu_0 NI}{2R}$. $\mu_0 = 4\pi \times 10^{-7}$, $N = 500$, $I = 2$,

$R = 0.1$ m.

$B = \frac{4\pi \times 10^{-7} \times 500 \times 2}{2 \times 0.1} = \frac{4\pi \times 10^{-4}}{0.2} = 2\pi \times 10^{-3}$ T.

(NCERT Ed. 2025-26, 12th, Page No. 116)

30. (1)

In a closed pipe, frequencies are odd harmonics $(f, 3f, 5f \ldots)$.

Fundamental = $f_1 = \frac{v}{4L}$.

1st Overtone = 3rd Harmonic = $\frac{3v}{4L}$.

$220 = \frac{3 \times 330}{4L}$

$L = \frac{990}{880} = \frac{9}{8} = 1.125$ m.

(NCERT Ed. 2025-26, 11th, Page No. 293)

31. (1)

Statement I: Correct. Fringe width $\beta = \frac{\lambda D}{d}$. Since $\lambda_{violet} < \lambda_{yellow}$, the fringe width decreases.

Statement II: Correct. Position of 1st minima $y_1 = \frac{\lambda D}{2d} = \frac{\beta}{2}$. The distance between them is $\frac{1}{2}(\beta_{yellow} - \beta_{violet})$.

(NCERT Ed. 2025-26, 12th, Page No. 266)

32. (3)

Differentiating (Chain Rule):

$\frac{d\phi}{dt} = 2(t^2 + 3t) \times (2t + 3)$

At $t = 1$ s:

$|e| = 2(1^2 + 3(1)) \times (2(1) + 3)$

$|e| = 2(4) \times (5) = 40$ mV

(NCERT Ed. 2025-26, 12th, Page No. 158)

33. (3)

Brewster's Law: $\tan i_p = \mu$.

$\tan i_p = \sqrt{3}$.

$i_p = 60^\circ$.

34. (4)

Using $\frac{1}{f} = (\mu_g - 1)\frac{2}{R}$:

$\frac{1}{f} = (1.5 - 1)\frac{2}{R} = \frac{1}{R} \Rightarrow R = f$

Two Plano-convex glass halves (f_1, f_3):

$\frac{1}{f_1} = \frac{1}{f_3} = (\mu_g - 1)\frac{1}{R} = \frac{0.5}{f} = \frac{1}{2f}$

Biconcave liquid layer (f_2):

$\frac{1}{f_2} = (\mu_l - 1)\left(-\frac{2}{R}\right) = (1.7 - 1)\left(-\frac{2}{f}\right) = -\frac{1.4}{f}$

$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$

$\frac{1}{f_{eq}} = \frac{1}{2f} - \frac{1.4}{f} + \frac{1}{2f} = \frac{1}{f} - \frac{1.4}{f} = -\frac{0.4}{f}$

$f_{eq} = \frac{-f}{0.4} = -2.5f$

(NCERT Ed. 2025-26, 12th, Page No. 238)

35. (4)

Power $P = V_{rms} I_{rms} \cos\phi$.

$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{100}{\sqrt{2}}$;

$I_{rms} = \frac{I_0}{\sqrt{2}} = \frac{100}{\sqrt{2}}$.

Phase difference $\phi = \frac{\pi}{3} = 60°$.

$P = \frac{100}{\sqrt{2}} \times \frac{100}{\sqrt{2}} \times \cos(60°)$

$= \frac{10000}{2} \times \frac{1}{2} = \frac{10000}{4} = 2500$ W.

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[Contd...

36. (3)

- $E_{g1} = 0$ eV: Characteristic of a Metal (Conductor).
- $E_{g2} = 1.1$ eV : (≤ 3 eV) Characteristic of a Semiconductor (like Silicon).
- $E_{g3} = 5.4$ eV : (> 3 eV) Characteristic of an Insulator (like Diamond).

(NCERT Ed. 2025-26, 12th, Page No. 326)

37. (1)

1 atomic mass unit (u) is equivalent to 931.5 MeV energy.

Energy $E = \Delta m \times 931.5$.

$E = 0.03 \times 931.5 \approx 27.945$ MeV.

Closest option is 28 MeV.

(NCERT Ed. 2025-26, 12th, Page No. 311)

38. (2)

The Balmer series results from electronic transitions to the $n = 2$ state. Its primary lines (from $n = 3, 4, 5, 6$) produce wavelengths between 400 nm and 700 nm (visible light), while the Lyman series is entirely in the Ultraviolet and others are in the Infrared region.

(NCERT Ed. 2025-26, 12th, Page No. 297)

39. (3)

A charge at rest produces only an electric field. A charge moving with constant velocity produces magnetic and electric fields but no radiation. An accelerated charge produces oscillating electric and magnetic fields, which regenerate each other and propagate as electromagnetic waves.

(NCERT Ed. 2025-26, 12th, Page No. 205)

40. (3)

The negative slope of graph between V and I should give internal resistance.

$$\Rightarrow r = \frac{5}{20} = 0.25\ \Omega$$

(NCERT Ed. 2025-26, 12th, Page No. 94)

41. (3)

As $a = v\dfrac{dv}{dx}$

$a = 15(5) = 75$ m/s^2

(NCERT Ed. 2025-26, 11th, Page No. 15)

42. (4)

Process AB is isochoric $\Rightarrow W_1 = $ zero

Process BC is isothermal

$$\Rightarrow W_2 = nR(4T_0)\ln\left(\frac{4V_0}{2V_0}\right) = 8RT_0\ln(2)$$

Total work done by gas $= W_1 + W_2 = 8RT_0\ln(2)$

(NCERT Ed. 2025-26, 11th, Page No. 234)

43. (3)

$\Rightarrow \lambda = 2(l_2 - l_1)$

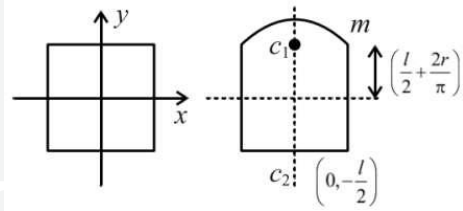
$= 2(47 - 15)$

$= 2 \times 32$

$= 64$ cm

(NCERT Ed. 2025-26, 11th, Page No. 292)

44. (1)



$y_{cm} = 0$

$$y'_{cm} = \frac{m\left(\frac{l}{2} + \frac{l}{\pi}\right) - \frac{ml}{2}}{4m} = \frac{l}{4\pi}$$

$\therefore$ shift $= \dfrac{l}{4\pi}$

(NCERT Ed. 2025-26, 11th, Page No. 96)

45. (2)

Tension (T) depends on the direction of acceleration (a):

- A-IV: Accelerating up (Force required > Weight): $T = m(g + a) = 40(10 + 2) = 480$ N.
- B-I: Accelerating down (Force required < Weight): $T = m(g - a) = 40(10 - 2) = 320$ N.
- C-II: Constant speed means zero acceleration ($a = 0$): $T = mg = 40(10) = 400$ N.
- D-III: Sliding freely means free fall ($a = g$ downwards): $T = m(g - g) = 0$ N.

(NCERT Ed. 2025-26, 11th, Page No. 65)

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46. (1)

Solution having the same osmotic pressure is called isotonic solutions.

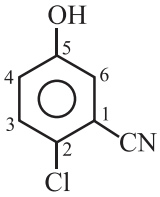
$\pi_{\text{cane sugar}} = \pi_{\text{(unknown solute)}}$

$$\frac{5.12}{342} = \frac{0.9}{M}$$

$$M = \frac{342 \times 0.9}{5.12} = 60$$

(NCERT Ed. 2025-26, 12th, Page No. 22)

47. (4)

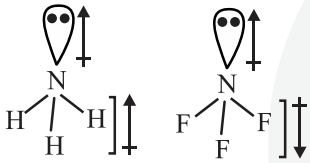


2-chloro-5-hydroxybenzonitrile

(NCERT Ed. 2025-26, 11th, Page No. 266)

48. (3)

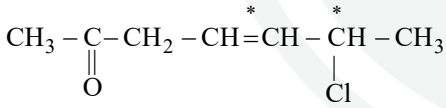
$BF_3 \Rightarrow AB_3L_0 \Rightarrow \mu = 0$



$\mu = 1.47$ D $\quad \mu = 0.23$ D

(NCERT Ed. 2025-26, 11th, Page No. 112)

49. (1)



Stereoisomer includes both optical and geometrical isomer.

Compound is Asymmetric so total number of Stereoisomers = $2^n = 2^2 = 4$

(NCERT Ed. 2025-26, 12th, Page No. 19)

50. (3)



, $PhNH_2$, CH_3NH_2, $C_6H_{11}–NH_2$,

These 4 can only be prepared by Gabriel phthalimide

(NCERT Ed. 2025-26, 12th, Page No. 264)

51. (2)

For MnS and ZnS, $K_{sp} = s^2$

For Bi_2S_3 $K_{sp} = 108s^5$

For Ag_2S $K_{sp} = 4s^3$

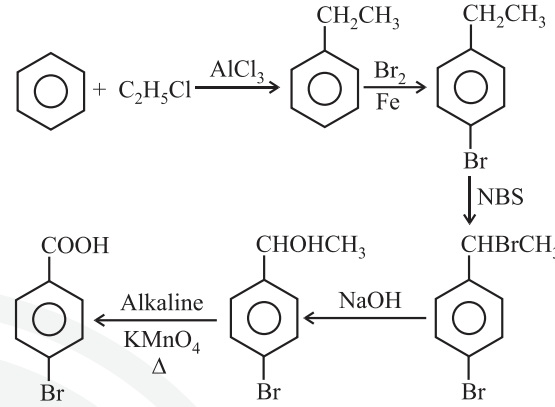
(NCERT Ed. 2025-26, 11th, Page No. 205)

52. (4)

$$\frac{1}{\lambda} = Rz^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), n_2 = 2, n_1 = 1$$

(NCERT Ed. 2025-26, 11th, Page No. 48)

53. (4)



(NCERT Ed. 2025-26, 12th, Page No. 20)

54. (1)

From Avogadro's law :

$\because$ 3.011×10^{22} atoms contain an element weight 1.15 gm.

Atomic mass $\rightarrow$ 1 mole of atoms $\rightarrow 6.022 \times 10^{23}$ atoms

$$1 \text{ atom} = \frac{1.15}{3.011 \times 10^{22}}$$

$$6.022 \times 10^{23} \text{ atoms} = \frac{1.15 \times 6.022 \times 10^{23}}{3.011 \times 10^{22}} = 23$$

Thus, the atomic mass of the element is = 23.

(NCERT Ed. 2025-26, 11th, Page No. 18)

55. (1)

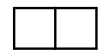
In the given equilibrium reaction if an inert gas is added at a constant pressure it will result in increased volume due to which the equilibrium shift in the direction where no. of gaseous moles are more, so option (1) is correct.

(NCERT Ed. 2025-26, 11th, Page No. 187)

56. (4)

For tetrahedral complex





Hence $(d_{x^2-y^2} = d_{z^2}) < (d_{xy} = d_{yz} = d_{xz})$

(NCERT Ed. 2025-26, 12th, Page No. 133)

[Contd...

57. (3)

In Fe_3O_4, the average oxidation number of Fe is:

$$\frac{(+2)+2(+3)}{3}=\frac{8}{3}$$

This appears as a fractional oxidation state (+8/3).

Fe_3O_4 contains both Fe^{2+} and Fe^{3+} ions.

Actual composition:

$$Fe_3O_4 = FeO \cdot Fe_2O_3$$

(NCERT Ed. 2025-26, 11th, Page No. 11)

58. (2)

$$t=\frac{2.303}{10^{-3}}\log\left(\frac{100}{25}\right)$$

$$t = 1386.4 \text{ sec}$$

(NCERT Ed. 2025-26, 12th, Page No. 13)

59. (4)

Enantiomers have the same:

- melting point
- boiling point
- density
- solubility (in achiral solvents)

But they differ in optical rotation:

- one rotates plane-polarized light clockwise (+)
- the other anticlockwise (–).

(NCERT Ed. 2025-26, 12th, Page No. 19)

60. (2)

Specific conductivity (K)

$$=\frac{1}{R}\times \text{ Cell constant}$$

$$\text{Cell constant } = K \times R$$

$$= 0.0129 \times 100 = 1.29 \text{ cm}^{-1}$$

(NCERT Ed. 2025-26, 12th, Page No. 14)

61. (3)

For O_2 :

Bonding electrons = 10

Antibonding electrons = 6

$$BO=\frac{10-6}{2}=2$$

For O_2^- :

One extra electron enters an antibonding π* orbital

Antibonding electrons = 7

$$BO=\frac{10-7}{2}=1.5$$

The extra electron enters the π^*_{2p} antibonding molecular orbital.

(NCERT Ed. 2025-26, 11th, Page No. 131)

62. (1)

$$CH_3-CH=CH_2 \xrightarrow{NOCl} CH_3-\underset{Cl}{\underset{|}{CH}}-\underset{NO}{\underset{|}{CH_2}}$$

(NCERT Ed. 2025-26, 11th, Page No. 311)

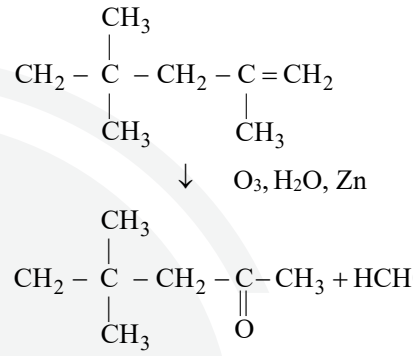
63. (2)

$$\Delta S_{vap}=\frac{\Delta H_{vap}}{T}=\frac{37.3\,kJ\,mol^{-1}}{373\,K}=0.1\ kJ\,mol^{-1}\,K^{-1}$$

$$=100\,J\,mol^{-1}\,K^{-1}$$

(NCERT Ed. 2025-26, 11th, Page No. 159)

64. (1)



(NCERT Ed. 2025-26, 11th, Page No. 313)

65. (4)

Given,

$W = 30\,g, I = 5\,A$

$W = Z \times I \times t$

Where,

Z = Electrochemical Equivalent

$$30=\frac{Eq\cdot wt}{96500}\times 5\times t$$

$$\text{Eq. wt}=\frac{40}{2}$$

$$30=\frac{40}{2\times 96500}\times 5\times t$$

$$t=\frac{30\times 2\times 96500}{40\times 5}$$

$$t = 289505 \text{ second}$$

$$=\frac{28950}{3600}\approx 8 \text{ hours}$$

(NCERT Ed. 2025-26, 12th, Page No. 22)

66. (1)

3s	2
3p	1
3d	0
4s	3

(Radial nodes = $n - l - 1$)

(NCERT Ed. 2025-26, 11th, Page No. 59)

[Contd...

67. (2)

Given reaction A → B

$\Delta H = 20\ kJ\ mol^{-1}$

Energy of activation for forward reaction

$(E_a) = 85$ kJ/mol and $\Delta H = 20$ KJ/mol

∴ Energy of activation for backward reaction

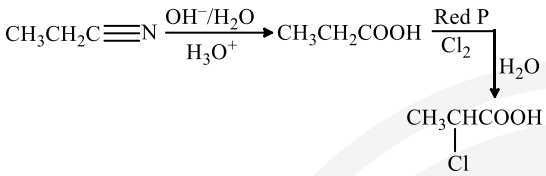
$E_b = E_a - \Delta H$

$= 85 - 20$

$(E_b) = 65$ KJ/mole.

(NCERT Ed. 2025-26, 12th, Page No. 19)

68. (1)



(NCERT Ed. 2025-26, 12th, Page No. 20)

69. (1)

$V_2 = 2V_1$	nR ln 2
$V_2 = 3V_1$	nR ln 3
$V_2 = V_1$	0
$V_2 = eV_1$	nR

$\left(\Delta S = nR \ln \frac{V_2}{V_1}\right)$

(NCERT Ed. 2025-26, 12th, Page No. 253)

70. (3)

For effective covalent bond formation, orbitals should have:

- Comparable energies
- Proper orientation
- Significant overlap

Example: 1s-1s overlap in H_2 or 2p-2p overlap in O_2.

When orbital energies differ greatly, electron density cannot stabilize between the nuclei, so effective overlap and bond formation become difficult.

(NCERT Ed. 2025-26, 11th, Page No. 127)

71. (4)

H_2O - bent

XeO_4 - tetrahedral

ClO_4^- - tetrahedral

(NCERT Ed. 2025-26, 11th, Page No. 114)

72. (1)

NaCl	2
$CaCl_2$	3
Glucose	1
$AlCl_3$	4

(NCERT Ed. 2025-26, 12th, Page No. 25)

73. (4)

Rate = $k[A]^x$

If $[A] = 2.4 \times 10^{-2}$ then rate = r

$r = k\ [2.4 \times 10^{-2}]^x$ …(i)

IF $[A] = 1.2 \times 10^{-2}$ then rate = $\frac{r}{8}$

$\frac{r}{8} = k\left[1.2 \times 10^{-2}\right]^x$ …(ii)

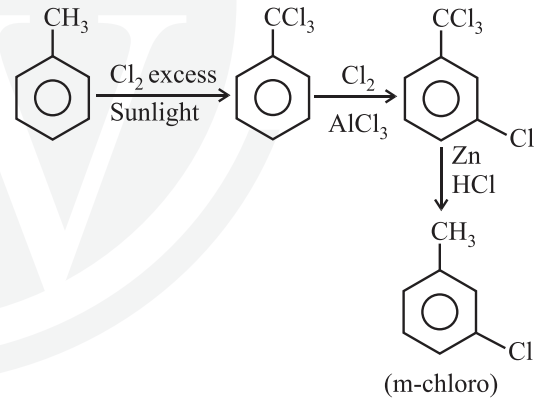
on dividing eq^n (i) by eq^n (ii)

$\frac{r}{\frac{r}{8}} = (2)^x$

$8 = (2)^x = x = 3$

(NCERT Ed. 2025-26, 12th, Page No. 8)

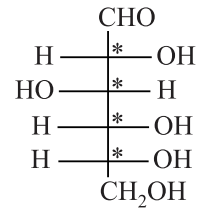
74. (1)



(NCERT Ed. 2025-26, 11th, Page No. 325)

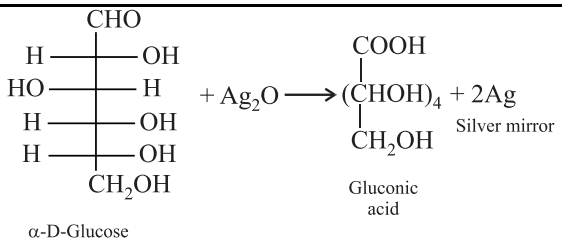
75. (3)

The chemical formula of glucose is $C_6H_{12}O_6$. The structure can be written as-



Glucose have four asymmetric centre and its gets mutarotation. It reacts with the hydroxyl amine but when glucose react with Tollen's reagent then glucoxic acid is formed.

[Contd...



(NCERT Ed. 2025-26, 12th, Page No. 5)

76. (2)

Given that, T = 400 K, V = 1 L, P = 600 mm Hg In reversible reaction -

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

Initial mole	100	0
At equilibrium	(100 – 20) =80	40

∴ Total number of moles = 40 + 80 = 120

$$\therefore \quad K_p = \frac{\left(p_{NO_2}\right)^2}{p_{N_2O_4}}$$

$$K_p = \frac{\left(\frac{40}{120} \times 600\right)^2}{\frac{80}{120} \times 600}$$

$$K_p = 100$$

(NCERT Ed. 2025-26, 11th, Page No. 183)

77. (3)

O_2	2
O_2^-	1.5
N_2^-	2.5
B_2	1

(NCERT Ed. 2025-26, 11th, Page No. 131)

78. (1)

Definition of adiabatic process:

q = 0

No heat is exchanged between the system and surroundings.

In an adiabatic process, temperature usually changes because work is done.

Examples:

- Adiabatic expansion → temperature decreases
- Adiabatic compression → temperature increases

Constant temperature occurs in isothermal processes, not adiabatic ones.

(NCERT Ed. 2025-26, 11th, Page No. 138)

79. (4)

It is pK_b of ammonia solution, $pK_b = 4.75$. It is a buffer solution.

$$pOH = pK_b + \log\left(\frac{\text{salt}}{\text{base}}\right)$$

$$pOH = 4.75 + \log\frac{0.2}{0.1}$$

$$pOH = 4.75 + \log 2$$

$$pOH = 4.75 + 0.30$$

$$pOH = 5.05$$

$$pH = 14 - pOH$$

$$pH = 14 - 5.05$$

$$pH = 8.95$$

(NCERT Ed. 2025-26, 11th, Page No. 198)

80. (3)

In the process denaturation, secondary and tertiary structures of protein get destroyed but the primary structure remains undistributed. Since denatureation reactions are not strong enough to break the peptide bonds, the primary structure (sequence of amino acids) remains the same after a denaturation process.

(NCERT Ed. 2025-26, 12th, Page No. 14)

81. (4)

Given that,

$$\lambda_A = 33.3\,\text{nm or } 33.33 \times 10^{-9}\,\text{m}$$

$$P_B = \frac{1}{3}P_A$$

According to de-Broglie-

$$\lambda = \frac{h}{p}$$

$$\lambda_A = \frac{h}{P_A} \qquad \text{.....(i)}$$

$$\lambda_B = \frac{h}{\frac{P_A}{3}} = \frac{3h}{P_A} \qquad \text{....(ii)}$$

From equation (i) and (ii), we get

$$\frac{\lambda_A}{\lambda_B} = \frac{1}{3} \Rightarrow \lambda_B = 33.33 \times 10^{-9} \times 3 = 99.99 \times 10^{-9}\,\text{m}$$

$$= 1.0 \times 10^{-7}\,\text{m}$$

(NCERT Ed. 2025-26, 11th, Page No. 50)

82. (3)

$$\Delta T_b = 2 \times K_b \times m$$

For KCl (i = 2)

$$= 2 \times 0.52 \times 1 = 1.04$$

$$T_b = T_b^\circ + \Delta T_b = 100 + 1.04 = 101.04^\circ C$$

(NCERT Ed. 2025-26, 12th, Page No. 24)

[Contd...

83. (1)

This is Faraday's First Law of Electrolysis:

$m \propto Q$

or

$m = ZQ$

where

- m = mass deposited
- Q = charge passed
- Z = electrochemical equivalent

According to Faraday's Second Law:

1 Faraday = 96500 C

This amount of charge deposits one gram equivalent of a substance, not one gram.

(NCERT Ed. 2025-26, 12th, Page No. 21)

84. (3)

Along a period first ionization energy increases. Thus, the first IE of the elements in second period should follow the order Be < B < N < O

The first IE of these elements follows the order B < Be < O < N The lower IE of B than that of Be is because in B ($1s^2$, $2s^2$ $2p^1$), electron is to be removed from 2p which is easy while in Be ($1s^2$, $2s^2$), electron is to be removed from 2s which is difficult. The low IE of O than that of N is because of the half-filled 2 p orbitals in N ($1s^2$, $2s^2$ $2p^3$).

(NCERT Ed. 2025-26, 11th, Page No. 89)

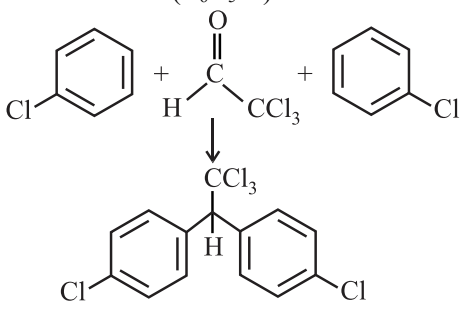
85. (2)

- The order of a reaction is determined experimentally from the rate law.
- It may or may not equal the stoichiometric coefficients of the balanced equation.
- Equality occurs only for elementary reactions, not for complex reactions.
- Reaction mechanism determines the rate law.
- Since order comes from the rate law, it depends on the reaction mechanism.

(NCERT Ed. 2025-26, 12th, Page No. 8)

86. (2)

DDT (Dichlorodiphenyltrichloroethane) is obtained by the reaction of chloral (CCl_3CHO) with chlorobenzene (C_6H_5Cl). DDT is an insecticide.



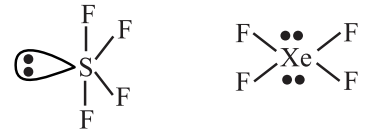
DDT

(NCERT Ed. 2025-26, 12th, Page No. 30)

87. (4)

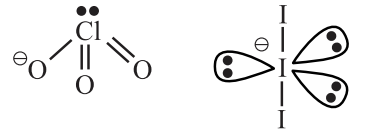
Structure of given compound is –

SF_4 XeF_4



1 lone pair 2 lone pair

ClO_3^- I_3^-



1 lone pair 3 lone pair

So the option (4) is correct.

(NCERT Ed. 2022-23, 12th, Page No. 210)

88. (2)

0.4 moles of cyclohexane

No. of atoms = $0.4 \times 18 = 7.2\ N_A$

68.4g sucrose

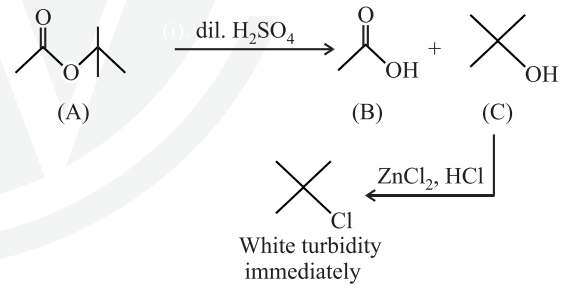
No. of atoms = $0.2 \times 45 = 9\ N_A$

70. 70.2L of oxygen

No. of atom = $3.13 \times 2 = 6.26\ N_A$

(NCERT Ed. 2025-26, 11th, Page No. 18)

89. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 208)

90. (2)

Given that,

$BaCl_2 = 280 \times 10^{-4}$

$NaOH = 248 \times 10^{-4}$

$NaCl = 126 \times 10^{-4}$

$BaCl_2 + 2NaOH \rightarrow Ba(OH)_2 + 2NaCl$

$$\lambda^{\infty}_{m\,Ba(OH)_2} = \lambda^{\infty}_{m\,BaCl_2} + 2\lambda^{\infty}_{m\,NaOH} - 2\lambda^{\infty}_{m\,NaCl}$$

$$= 280\times10^{-4} + 2\times248\times10^{-4} - 2\times126\times10^{-4}$$

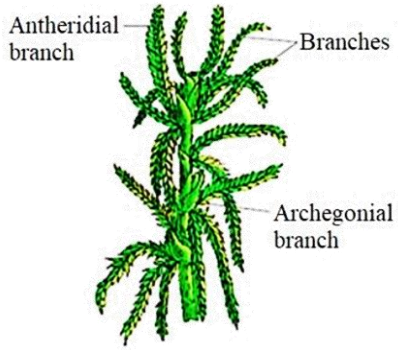
$$= (280 + 496 - 252)\times10^{-4}$$

$$= 524\times10^{-4}\,Sm^2\,mol^{-1}$$

(NCERT Ed. 2025-26, 12th, Page No. 20)

[Contd...

91. (2)



The plant shown in the given diagram represents *Sphagnum* gametophyte. It is a moss and doesn't produce seeds. Seeds are produced without seed coat in gymnosperms. In angiosperms, the seeds are enclosed in fruits.

(NCERT Ed. 2025-26, 11th, Page No. 28)

92. (4)

The whole scheme of transfer of electrons, starting from the PS II, uphill to the acceptor, down the electron transport chain to PS I, excitation of electrons, transfer to another acceptor, and finally down hill to $NADP^+$ reducing it to $NADPH + H^+$ is called the Z scheme, due to its characterstic shape. The Z scheme (non-cyclic photophosphorylation) occurs in the grana thylakoids. The stroma lamellae lack Photosystem II (PS II) and the NADP reductase enzyme, so only cyclic photophosphorylation occurs there. The characteristic "Z" shape is formed when the electron carriers are placed in sequence according to their redox potential (oxidation-reduction potential), not a light intensity scale. The Z scheme involves both PS II and PS I working in series. This non-cyclic flow of electrons results in the photolysis of water (releasing oxygen), the synthesis of ATP, and the reduction of $NADP^+$ to $NADPH + H^+$.

(NCERT Ed. 2025-26, 11th, Page No. 139)

93. (1)

Since resources for growth for most animal populations are finite and become limiting sooner or later, the logistic growth model is considered a more realistic one.

(NCERT Ed. 2025-26, 12th, Page No. 196)

94. (1)

Monera	:	Cell wall non-cellulosic (polysaccharide + amino acid)
Protista	:	Cell type eukaryotic, Cell wall present in some
Fungi	:	Cell wall present with chitin
Animalia	:	Body organisation tissue/ organ/ organ system

(NCERT Ed. 2025-26, 11th, Page No. 11)

95. (1)

Meiosis I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the S phase. Meiosis involves pairing of homologous chromosomes and recombination between non-sister chromatids of homologous chromosomes. Four haploid cells are formed at the end of meiosis II.

(NCERT Ed. 2025-26, 11th, Page No. 125)

96. (2)

The inheritance of flower colour in the dog flower (snapdragon or *Antirrhinum* sp.) is a good example to understand incomplete dominance.

(NCERT Ed. 2025-26, 12th, Page No. 60)

97. (2)

The decomposition is quicker if detritus is rich in nitrogen/sugars and if the environment is warm and moist. Setup I + Detritus B provides the ideal combination of "quality" and "climate." Option 1 is incorrect because lignin significantly slows decomposition even in good climates. Option 3 and 4 is incorrect because anaerobiosis and low temperature inhibit decomposition (leading to buildup, not necessarily faster processing into humus).

(NCERT Ed. 2025-26, 12th, Page No. 208)

98. (3)

The final stage of meiotic prophase I is diakinesis. This is marked by terminalisation of chiasmata. During this phase the chromosomes are fully condensed and the meiotic spindle is assembled to prepare the homologous chromosomes for separation.

(NCERT Ed. 2025-26, 11th, Page No. 126)

99. (4)

Steps of DNA fingerprinting:

(i) isolation of DNA,
(ii) digestion of DNA by restriction endonucleases,
(iii) separation of DNA fragments by electrophoresis,
(iv) transferring (southern blotting) of separated DNA fragments to synthetic membranes, such as nitrocellulose or nylon,
(v) hybridisation using labelled VNTR probe, and
(vi) detection of hybridised DNA fragments by autoradiography.

(NCERT Ed. 2025-26, 12th, Page No. 106)

100. (4)

Agaricus belongs to basidiomycetes.

(NCERT Ed. 2025-26, 11th, Page No. 18)

101. (3)
RNA polymerase III is responsible for transcription of tRNA, 5srRNA and snRNAs in eukaryotes. tRNA is essential for reading the genetic code and bringing the correct amino acids to the ribosome during translation (synthesis of polypetides). If RNA polymerase III is defective, the cell cannot produce tRNAs. Without tRNAs, the "translation machinery" cannot function, and the synthesis of polypeptides (proteins) in the cytoplasm will stop.
(NCERT Ed. 2025-26, 12th, Page No. 95, 99)

102. (2)
The water splitting complex is associated with the PS II, which itself is physically located on the inner side of the membrane of the thylakoid. The ATP synthase enzyme consists of two parts: one called the CF_0 is embedded in the thylakoid membrane and forms a transmembrane channel that carries out facilitated diffusion of protons across the membrane. The other portion is called CF_1 and protrudes on the outer surface of the thylakoid membrane on the side that faces the stroma.
(NCERT Ed. 2025-26, 11th, Page No. 142)

103. (4)
Morgan found that the genes white eye and yellow body were very tightly linked and showed only 1.3 per cent recombination while white eye and miniature wing showed 37.2 per cent recombination.
(NCERT Ed. 2025-26, 12th, Page No. 67)

104. (3)
The members of chlorophyceae are commonly called green algae. The plant body may be unicellular, colonial or filamentous. They are usually grass green due to the dominance of pigments chlorophyll *a* and *b*. The pigments are localised in definite chloroplasts.
(NCERT Ed. 2025-26, 11th, Page No. 26)

105. (1)
One single maize root apical meristem can give rise to more than 17,500 new cells per hour, whereas cells in a watermelon may increase in size by upto 3,50,000 times. In the former, growth is expressed as increase in cell number; the latter expresses growth as increase in size of the cell. While the growth of a pollen tube is measured in terms of its length, an increase in surface area denotes the growth in a dorsiventral leaf.
(NCERT Ed. 2025-26, 11th, Page No. 168)

106. (2)
The golgi apparatus principally performs the function of packaging materials, to be delivered either to the intra-cellular targets or secreted outside the cell. Many of the organelles are considered together as an endomembrane system because their functions are coordinated. The endomembrane system includes endoplasmic reticulum (ER), golgi complex, lysosomes and vacuoles.
(NCERT Ed. 2025-26, 11th, Page No. 96)

107. (1)
Ecology is one such thread which gives us a holistic perspective to biology. A particular aspect of this is the study of anthropogenic environmental degradation and the socio-political issues it has raised.
(NCERT Ed. 2025-26, 12th, Page No. 189)

108. (3)
In bacteria, since the mRNA does not require any processing to become active, and also since transcription and translation take place in the same compartment (there is no separation of cytosol and nucleus in bacteria), many times the translation can begin much before the mRNA is fully transcribed.
(NCERT Ed. 2025-26, 12th, Page No. 94)

109. (1)
Removing the shoot tip (source of Auxin) removes apical dominance, allowing lateral buds to grow. Cytokinins help overcome the apical dominance. Applying it to lateral buds will trigger growth even if the tip is present. ABA is a growth inhibitor and promotes dormancy; it would not promote lateral growth, rather it would further suppress activity. Water stress increases ABA levels (stress hormone), which leads to stomatal closure and growth inhibition, not lateral branching.
(NCERT Ed. 2025-26, 11th, Page No. 75, 76, 77)

110. (2)
In glycolysis, phosphoenol pyruvate (PEP) is converted into pyruvate (the final step). The primary precursor for fatty acid synthesis is Acetyl-CoA, not PEP.
(NCERT Ed. 2025-26, 11th, Page No. 146)

111. (1)
A small amount of curd added to the fresh milk as inoculum or starter contain millions of LAB, which at suitable temperatures multiply, thus converting milk to curd, which also improves its nutritional quality by increasing vitamin B_{12}.
(NCERT Ed. 2025-26, 12th, Page No. 151)

112. (3)
As per the given question, none of the progeny will have 'bb' genotype, all will have 'Bb'.
(NCERT Ed. 2025-26, 12th, Page No. 64)

113. (4)
Trees occupy top vertical strata or layer of a forest, shrubs the second and herbs and grasses occupy the bottom layers.
(NCERT Ed. 2025-26, 12th, Page No. 206)

[Contd...

114. (2)
During lactic acid fermentation, NADH + H^+ is oxidised to NAD^+.
(NCERT Ed. 2025-26, 11th, Page No. 157)

115. (1)

Oscillatoria	Biofertiliser
Nucleopolyhedrovirus	Bicontrol agent
Saccharomyces cerevisiae	Yeast used in making bread and beer
Monascus purpureus	Yeast, source of statins

(NCERT Ed. 2025-26, 12th, Page No. 153)

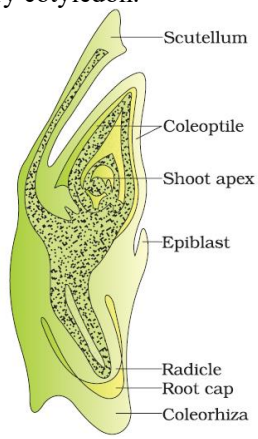
116. (4)
In coconut which is a drupe, the mesocarp is fibrous with a stony and hard endocarp.
(NCERT Ed. 2025-26, 11th, Page No. 66)

117. (1)
Colombia located near the equator has nearly 1,400 species of birds while New York at 41° N has 105 species and Greenland at 71° N only 56 species. India, with much of its land area in the tropical latitudes, has more than 1,200 species of birds.
(NCERT Ed. 2025-26, 12th, Page No. 219)

118. (4)
There are instances where a single gene can exhibit multiple phenotypic expression. Such a gene is called a pleiotropic gene. The 'I' gene of the ABO blood group system is the classic example of multiple allelism (having three alleles: I^A, I^B and i) and co-dominance (seen in the AB blood group). It is not considered a pleiotropic gene because it controls a single character: the type of sugar polymers (antigens) on the surface of red blood cells.
(NCERT Ed. 2025-26, 12th, Page No. 69)

119. (2)
The pith is small or inconspicuous in a dicot root.
(NCERT Ed. 2025-26, 11th, Page No. 74)

120. (2)
In the embryo of grass, the epiblast represents the rudimentary cotyledon.



L.S. of an embryo of grass
(NCERT Ed. 2025-26, 12th, Page No. 19)

121. (3)
In amensalism one species is harmed whereas the other is unaffected. Both the species benefit in mutualism.
(NCERT Ed. 2025-26, 12th, Page No. 197)

122. (4)
Evolutionarily, the pteridophytes are the first terrestrial plants to possess vascular tissues – xylem and phloem.
(NCERT Ed. 2025-26, 11th, Page No. 30)

123. (3)
At the onset of anaphase, each chromosome arranged at the metaphase plate is split simultaneously and the two daughter chromatids, now referred to as daughter chromosomes of the future daughter nuclei, begin their migration towards the two opposite poles. As each chromosome moves away from the equatorial plate, the centromere of each chromosome remains directed towards the pole and hence at the leading edge, with the arms of the chromosome trailing behind.
(NCERT Ed. 2025-26, 11th, Page No. 123)

124. (2)
There is a linear relationship between incident light and CO_2 fixation rates at low light intensities.
(NCERT Ed. 2025-26, 11th, Page No. 149)

125. (1)
In 1928, Frederick Griffith, in a series of experiments with *Streptococcus pneumoniae* witnessed a miraculous transformation in the bacteria. The unequivocal proof that DNA is the genetic material came from the experiments of Alfred Hershey and Martha Chase (1952). They worked with viruses that infect bacteria called bacteriophages. It was only in 1953 that James Watson and Francis Crick, based on the X-ray diffraction data proposed a very simple but famous Double Helix model for the structure of DNA. Very soon, Francis Crick proposed the Central dogma in molecular biology, which states that the genetic information flows from DNA→RNA→Protein.
(NCERT Ed. 2025-26, 12th, Page No. 84)

126. (3)
In dicot stems, the hypodermis is made of collenchyma; in monocot stems, it is made of sclerenchyma. It is generally not parenchymatous. The fruit wall (pericarp) of nuts is hard and woody because it is composed of sclerenchyma (specifically sclereids).
(NCERT Ed. 2022-23, 11th, Page No. 88)

[Contd...

127. (3)

Sporozoans: This includes diverse organisms that have an infectious spore-like stage in their life cycle. The most notorious is *Plasmodium* (malarial parasite) which causes malaria, a disease which has a staggering effect on human population.

(NCERT Ed. 2025-26, 11th, Page No. 16)

128. (2)

The vacuole is the membrane-bound space found in the cytoplasm. It contains water, sap, excretory product and other materials not useful for the cell. The vacuole is bound by a single membrane called tonoplast. In plants, the tonoplast facilitates the transport of a number of ions and other materials against concentration gradients into the vacuole.

(NCERT Ed. 2025-26, 11th, Page No. 96)

129. (4)

The 'rivet popper hypothesis' to explain the importance of species diversity in an ecosystem was proposed by Stanford ecologist Paul Ehrlich.

(NCERT Ed. 2025-26, 12th, Page No. 221)

130. (1)

Monkey, Gibbon and Human: All belong to the Order Primata.

Wheat (*Triticum aestivum*) belongs to the Order Poales.

Mango (*Mangifera indica*) belongs to the Order Sapindales.

Petunia belongs to the Order Polymoniales.

Dog belongs to the Order Carnivora.

(NCERT Ed. 2025-26, 11th, Page No. 7)

131. (2)

The process of formation of microspores from a pollen mother cell (PMC) through meiosis is called microsporogenesis. The PMC is a diploid (2n) cell located within the microsporangium (anther) of a flower. The PMC gives rise to the microspore that develops into the male gametophyte (pollen grain). The Megaspore Mother Cell (MMC) gives rise to the female gametophyte (embryo sac).

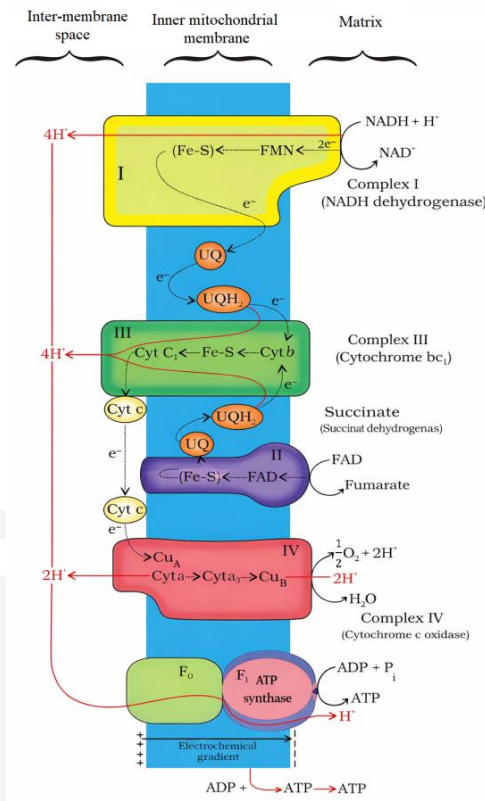
(NCERT Ed. 2025-26, 12th, Page No. 6)

132. (2)

Viruses are acellular (non-cellular) entities. They do not have a cell membrane or a cell wall. Instead, they consist of genetic material (RNA in the case of TMV) enclosed in a protein coat called a capsid.

(NCERT Ed. 2025-26, 11th, Page No. 20)

133. (4)



Electron Transport System (ETS)

(NCERT Ed. 2025-26, 11th, Page No. 160)

134. (1)

The meristem which occurs between mature tissues is known as intercalary meristem. They occur in grasses and regenerate parts removed by the grazing herbivores. Both apical meristems and intercalary meristems are primary meristems because they appear early in life of a plant and contribute to the formation of the primary plant body. The meristem that occurs in the mature regions of roots and shoots of many plants, particularly those that produce woody axis and appear later than primary meristem is called the secondary or lateral meristem. They are cylindrical meristems. Fascicular vascular cambium, interfascicular cambium and cork-cambium are examples of lateral meristems.

(NCERT Ed. 2022-23, 11th, Page No. 85)

135. (2)

Floral Formula of family Leguminosae:

% $K_{(5)}$ $C_{1+2+(2)}$ $A_{(9)+1}$ $\underline{G}_1$

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[Contd...

136. (3)

Ctenophores (comb jellies) possess eight rows of ciliated comb plates (ctenes) used for locomotion.

These structures are absent in cnidarians.

Ctenophores are diploblastic, not triploblastic, and they show tissue level of organisation, not organ-system level.

(NCERT Ed. 2025-26, 11th, Page No. 42)

137. (3)

The biosphere shows immense diversity—millions of species differing in structure, function, and habitat.

All the elements present in living tissues are also present in the earth's crust.

Biomolecules are defined as carbon compounds obtained from living tissues (proteins, carbohydrates, lipids, nucleic acids).

Coenzymes are organic cofactors that bind loosely and temporarily (transiently) to the apoenzyme during catalysis and then dissociate.

Adenosine, guanosine, uridine = nucleosides (base + sugar only).

Nucleotides = nucleoside + phosphate

(NCERT Ed. 2025-26, 11th, Page No. 118)

138. (2)

In a male cockroach, the genital pouch is a chamber that houses reproductive structures.

- Dorsal boundary → formed by 9th and 10th terga
- Ventral boundary → formed by 9th sternum
- It contains male gonapophyses and phallic structures

(NCERT Ed. 2020-21, 11th, Page No. 112)

139. (3)

- The epiglottis is a leaf-shaped cartilaginous flap located at the entrance of the larynx.
- During swallowing, it covers the glottis (laryngeal opening) to prevent food from entering the respiratory tract.
- If epiglottis is impaired:
- Food may enter the larynx/trachea → choking or aspiration
- Hence, the most directly affected function is protection of airway during swallowing

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140. (1)

List-I	List-II
Neutrophils	60-65 percent
Basophils	0.5-1 percent
Lymphocytes	20-25 percent
Monocytes	6-8 percent

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141. (3)

An increase in blood flow to the atria of the heart triggers the release of Atrial Natriuretic Factor (ANF). ANF causes vasodilation (dilation of blood vessels) and thereby decreases blood pressure, thus acting as a counter-regulatory check on the Renin-Angiotensin mechanism.

(NCERT Ed. 2025-26, 11th, Page No. 212)

142. (1)

Troponin is a key regulatory protein located on the thin (actin) filaments. It works alongside tropomyosin to control the interaction between actin and myosin; when calcium binds to it, it changes shape to initiate the contraction process. Other options like insulin and trypsin are hormones or digestive enzymes and do not have a structural or functional role in the contraction of muscle fibers.

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143. (2)

The initial depolarisation is caused by a rapid influx of Na^+. Within a fraction of a second, this is followed by a rise in permeability to K^+. The rapid efflux of K^+ ions diffuses outside the membrane, restoring the resting potential, a process known as repolarisation.

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144. (3)

List-I	List-II
Melatonin	Pineal gland
Thymosin	Thymus gland
Erythropoietin	Juxtaglomerular cells of kidney
Cholecystokinin	Gastro-intestinal tract

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145. (3)

After the birth of the baby, the placenta is expelled from the uterus. In human females, oogonia are formed only during fetal life.

- At birth → fixed number of primary oocytes present
- No new oogonia are formed after birth

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[Contd...

146. (3)

Periodic abstinence and coitus interruptus are natural/traditional methods that work on the principle of avoiding the physical meeting of gametes without employing any external physical barrier devices (like diaphragms) or chemical/ hormonal alterations.

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147. (2)

The similar anatomical structure of forelimbs across diverse mammals, adapted to different needs, indicates common ancestry. This phenomenon is known as homology, which is a result of divergent evolution.

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148. (3)

Microsporum is a genus of fungi (not a bacterium) responsible for causing ringworms. Pneumonia is caused by bacteria like *Streptococcus pneumoniae* and *Haemophilus influenzae.*

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149. (1)

For two pieces of DNA to ligate (stick together), their overhanging single-stranded ends—often called sticky ends—must be perfectly complementary (base-pair matching).

Let's look at the sticky ends each enzyme creates by cutting at the arrows:

1. RE1: Cuts between A and A. This leaves an overhang of AGCT.
2. RE2: Cuts between C and A. This also leaves an overhang of AGCT.
3. RE3: Cuts between G and A. This also leaves an overhang of AGCT.

Even though the recognition sequences are different for each enzyme, the specific sequence of the overhang is identical for all three:

RE1 overhang: 5'-AGCT...

RE2 overhang: 5'-AGCT...

RE3 overhang: 5'-AGCT...

Because the overhangs are identical, they are compatible. RE1-digested DNA will find a perfect base-pairing match with DNA cut by RE2 or RE3. Since all three results in the same AGCT sticky end, they can all ligate to one another.

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150. (3)

Transgenic mice are developed to test vaccine safety (like polio vaccine). For chemical safety (toxicity) testing, transgenic animals are made to carry genes that make them more sensitive to toxic substances than non-transgenic animals, not more resistant.

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151. (3)

The feature described—absence of an air bladder (necessitating constant swimming)—are characteristic hallmarks of cartilaginous fishes belonging to Class Chondrichthyes.

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152. (2)

In an exothermic reaction, the reaction releases energy, meaning the potential energy content of the product 'P' is at a lower level than that of the substrate 'S'.

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153. (3)

During aestivation and hibernation, respiration occurs through the skin (cutaneous respiration), not lungs.

In water, skin acts as respiratory organ via diffusion.

Lungs are elongated, pink, sac-like structures in the thoracic cavity.

When submerged, respiration is mainly cutaneous, not buccal.

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154. (4)

Functional Residual Capacity (FRC) includes the Expiratory Reserve Volume (ERV) and the Residual Volume (RV). Because RV is the volume of air remaining in the lungs even after a forcible expiration and cannot be exhaled, it cannot be measured by a standard spirometer. Consequently, any capacity including RV (like FRC and Total Lung Capacity) cannot be directly estimated by spirometry alone.

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155. (2)

The number of QRS complexes in one minute represents the heart rate, which is 80 beats/min. Cardiac Output = Heart Rate × Stroke Volume. Therefore, 80 beats/min × 70 mL/beat = 5600 mL/min.

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[Contd...

156. (4)

The kidneys receive about 1100–1200 mL of blood per minute (renal blood flow). This is approximately 20–25% (≈ 1/5) of the
cardiac output (blood pumped by each ventricle per minute).
Cardiac output ≈ 5 L/min 1/5 of 5 L ≈ 1–1.25 L (≈ 1100–1200 mL)

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157. (2)

According to the description of the skeletal system, each rib is a thin flat bone connected dorsally to the vertebral column. It has two articulation surfaces on its dorsal end and is hence called bicephalic.

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158. (3)

The limbic system is located in the inner parts of the cerebral hemispheres and includes structures such as the amygdala and hippocampus. It is mainly involved in emotions, behavior, motivation, and memory.

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159. (3)

Parathyroid hormone (PTH) increases blood calcium levels (hypercalcemic hormone) by acting on bones to stimulate resorption and by stimulating reabsorption in renal tubules and absorption from digested food. Thyrocalcitonin (TCT) decreases blood calcium levels.

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160. (3)

After fertilisation, the zygote undergoes mitotic divisions (cleavage) as it moves through the oviduct (isthmus) toward the uterus, forming 2, 4, 8, 16 blastomeres.
The 8–16 cell stage is called morula, not blastula.
Later, morula develops into blastocyst (in humans, blastula stage is termed blastocyst).

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161. (3)

Amniocentesis is a diagnostic procedure used to test for genetic disorders, but it was being widely misused to determine the sex of the unborn child, frequently leading to illegal female foeticide. The statutory ban was enacted to legally check this menace.

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162. (3)

A. I. Oparin and J. B. S. Haldane proposed that life originated from pre-existing non-living organic molecules (chemical evolution).
Stanley L. Miller (1953) did not prove spontaneous generation. His experiment only showed that simple organic molecules (like amino acids) can form under primitive Earth conditions.

(NCERT Ed. 2025-26, 12th, Page No. 111)

163. (3)

Allergies are an exaggerated response of the immune system to certain antigens. The antibodies produced to these are of the IgE type. The symptoms are due to the release of chemicals like histamine and serotonin from the mast cells.

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164. (4)

List-I	List-II
ori	Initiates replication and controlling copy number.
Selectable marker	Usually an antibiotic resistance gene used to identify transformants.
Cloning site	Specific recognition sequence where restriction enzymes cut to link alien DNA.
Insertional inactivation	Loss of an enzyme's activity due to the insertion of alien DNA.

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165. (3)

In mammals, insulin is synthesised as a pro-hormone containing an extra stretch called the C peptide. This C peptide is not present in mature, functional insulin and is removed during maturation. Assembling insulin into this mature form was the main challenge in rDNA techniques.

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166. (4)

Annelids are bilaterally symmetrical and are true coelomates. Ctenophores and Coelenterates have radial symmetry. Platyhelminthes are bilaterally symmetrical but are acoelomates.

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167. (2)

The zwitterionic form of an amino acid occurs when it simultaneously carries a positively charged protonated amino group and a negatively charged deprotonated carboxyl group, typically at neutral pH.

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[Contd...

168. (3)

The nervous system of a cockroach is spread throughout the body. The head holds a bit of a nervous system (supra-oesophageal ganglion) while the rest is situated along the ventral (belly-side) part of its body as segmentally arranged ganglia.

(NCERT Ed. 2020-21, 11th, Page No. 114)

169. (2)

At the alveolar site where pCO_2 is low, the reaction proceeds in the opposite direction leading to the formation of CO_2 and H_2O from bicarbonate. This allows the CO_2 trapped as bicarbonate at the tissue level to be released out.

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170. (3)

Heart failure means the state of the heart when it is not pumping blood effectively enough to meet the body's needs. Cardiac arrest is when the heart stops beating, and a heart attack is when the heart muscle is suddenly damaged by an inadequate blood supply.

(NCERT Ed. 2025-26, 11th, Page No. 203)

171. (3)

List-I	List-II
Micturition	Nervous system control
Diabetes mellitus	Glucose in urine
Kidney failure	Hemodialysis
Sebaceous glands	Eliminate sterols

(NCERT Ed. 2025-26, 11th, Page No. 213)

172. (2)

8th, 9th and 10th ribs are false ribs; they do not articulate directly with the sternum, but join the 7th rib via costal cartilage.

The thumb (carpometacarpal) joint is a saddle joint, allowing opposable movement.

Myasthenia gravis is an autoimmune disorder affecting neuromuscular junction, causing fatigue and skeletal muscle weakness.

Acromion process is part of the scapula, not the clavicle.

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173. (1)

In a living cell, DNA helicase slowly unwinds the DNA, creating a replication fork. Because DNA polymerase can only build in the 5' to 3' direction, one strand (lagging strand) must be built in short, discontinuous chunks (Okazaki fragments) as the fork opens. In PCR, the denaturation step uses high heat to completely separate the two DNA strands along their entire length at once. Because the strands are fully separated and primers bind to the 3' end of each, *Taq* polymerase can continuously extend the new strands in the 5' to 3' direction on both templates without needing to wait for a moving fork. Since there is no replication fork, no lagging strand forms, so discontinuous synthesis does not occur.

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174. (4)

TSH (Thyroid Stimulating Hormone) does not act directly on body tissues. It is a tropic hormone that stimulates another endocrine gland (thyroid gland). TSH stimulates the thyroid gland to secrete T_3 (triiodothyronine) and T_4 (thyroxine), which then regulate metabolism.

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175. (4)

List-I	List-II
Menarche	The first menstruation beginning at puberty.
Spermiation	Release of mature sperms into lumen of seminiferous tubules
Colostrum	Milk produced during initial days of lactation.
Parturition	The process of childbirth

(NCERT Ed. 2025-26, 12th, Page No. 38)

176. (1)

The process of removing non-coding regions (introns) from a primary RNA transcript (hnRNA) to form mature mRNA is a defining characteristic of eukaryotic cells. Malaria is caused by *Plasmodium* (a sporozoan, protest, eukaryotic). Pertussis is caused by the bacterium *Bordetella pertussis*. Typhoid is caused by the bacterium *Salmonella typhi*. Tuberculosis is caused by the bacterium *Mycobacterium tuberculosis*.

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177. (1)

- $\rightarrow q^2 = \frac{4}{10000} = 0.0004$
- So, $q = \sqrt{0.0004} = 0.02$
- $\rightarrow p = 1 - q = 0.98$
- Heterozygous frequency = 2pq
- $\rightarrow 2 \times 0.98 \times 0.02 = 0.0392$
- Number of heterozygotes = 0.0392×10000 = 392 Hence, 392 individuals are heterozygous.

(NCERT Ed. 2025-26, 12th, Page No. 121)

[Contd...

178. (3)

- HIV infects macrophages and $CD4^+$ T-cells.
- Inside these cells, viral RNA is reverse-transcribed into DNA by reverse transcriptase, producing new virions, effectively turning the cell into an HIV "factory."
- The severe immunodeficiency in AIDS is mainly due to the progressive loss of $CD4^+$ T-lymphocytes, not B-lymphocytes.
- B-cells may be affected indirectly, but the primary defect is in helper T-cells, which orchestrate immune responses.

(NCERT Ed. 2025-26, 12th, Page No. 140)

179. (2)

Plasmids are autonomously replicating, circular, extrachromosomal DNA.

Bacteriophages, because of their high number per cell, have very high copy numbers of their genome within the bacterial cells as compared to the plasmids.

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180. (2)

The RNAi strategy involves introducing nematode-specific genes using *Agrobacterium* vectors that produce both sense and anti-sense RNA. These form double-stranded RNA (dsRNA) which initiates RNAi, silencing the specific mRNA of the nematode and causing its death.

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